

Level 1: Easy

1. Write a Python program that prints your name, age, and favorite color on separate lines.

Input format

There is no external input for this program.

You provide your name, age, and favorite color directly in the program as variables.

Output format

Print your name, age and favorite color on separate lines

Constraints

The name should be a string with a maximum length of 255 characters.

The age should be an integer between 0 and 150 (realistic age range).

The favorite color should be a string with a maximum length of 255 characters.

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:	Test Case 3:	Test Case 4:
Input	name = "John", age = 30, favorite_color = "Green"	name = "Alice", `age = 28`, `favorite_color = "Purple"	name = "Bob", `age = 42`, `favorite_color = "Red"	name = "Eva", `age = 55`, `favorite_color = "Yellow"
Output	Name: John Age: 30 Favorite Color: Green	Name: Alice Age: 28 Favorite Color: Purple	Name: Bob Age: 42 Favorite Color: Red	Name: Eva Age: 55 Favorite Color: Yellow

2. Create a Python program that calculates and prints the area of a circle.

Input format

The first line of each test case consists of an integer as N.

Output format

Print the area of a circle.

Constraints

The radius should be a positive floating-point number.

The program assumes that the user will enter a valid radius value.

The program will calculate and print the area without any specific limitations on the radius value as long as it's a valid positive number.

The program uses an approximate value of π (pi) for the calculation, and this approximation may not be as accurate as using the math module.

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:	Test Case 3:	Test Case 4:
Input	5.0	3.5	10.0	1.414
Output	The area of the circle with radius 5.0 is: 78.53975	The area of the circle with radius 3.5 is: 38.4845	The area of the circle with radius 10.0 is: 314.159	The area of the circle with radius 1.414 is: 6.276523

3. Write a Python program that takes two numbers as input and prints their sum.

Input format

The input consists of two integers.

Output format

Prints sum of the two input integers.

Constraints

Both input numbers should be integers.

The program assumes that the user will enter valid integer values.

The program will calculate and print the sum without any specific limitations on the input integers, except for their valid integer values.

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:	Test Case 3:	Test Case 4:
Input	Enter the first integer: 5 Enter the second integer: 3	Enter the first integer: -2 Enter the second integer: 7	Enter the first integer: 0 Enter the second integer: 0	Enter the first integer: 100 Enter the second integer: -50
Output	The sum of 5 and 3 is: 8	The sum of -2 and 7 is: 5	The sum of 0 and 0 is: 0	The sum of 100 and -50 is: 50

4. Implement a program that takes a user's name as input and greets them with "Hello, [Name]!"

Input format

The input consists of a single line where the user is prompted to enter their name.

Output format

Print the user's name as input and greets them with "Hello,[Name]!"

Constraints

The name entered by the user can be any valid string, including letters, numbers, and special characters.

The program assumes that the user will provide a name without any specific limitations or constraints on the length or character content.

The program will greet the user with the provided name without restrictions.

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:	Test Case 3:	Test Case 4:
Input	Enter your name: John	Enter your name: Alice	Enter your name:123	Enter your name:\$user
Output	Hello, John!	Hello, Alice!	Hello, 123!	Hello, \$user!

5. Implement a program that takes a user's birthdate as input and calculates their age.

Input format

The user is prompted to enter their birthdate as a string in the format "YYYY-MM-DD," where:

"YYYY" represents the year with four digits (e.g., 1990).

"MM" represents the month with two digits (e.g., 05 for May).

"DD" represents the day with two digits (e.g., 15).

Output format

The program displays the user's age as an integer in years.

Constraints

The program assumes that the user will provide a valid birthdate in the specified format.

The program calculates the age accurately based on the provided birthdate.

The age calculation takes into account leap years for accurate results.

The program does not handle future birthdates (birthdates beyond the current date) for age calculation. It assumes that the birthdate is in the past.

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:	Test Case 3:	Test Case 4:
Input	"1990-05-15"	"2000-12-20"	"1985-03-10"	"2015-08-25"
Output	"Your age is: 31 years"	"Your age is: 20 years"	"Your age is: 36 years"	"Your age is: 6 years"

Level 2: Medium

1. Build a program that requests the user's weight (in kilograms) and height (in meters) to calculate their body mass index (BMI)

Input format

The program takes two inputs from the user:

Weight in kilograms (a positive floating-point number).

Height in meters (a positive floating-point number).

Output format

The program calculates the Body Mass Index (BMI), which is a floating-point number representing the BMI value.

Constraints

The weight input must be a positive floating-point number (greater than 0).

The height input must be a positive floating-point number (greater than 0).

The program does not handle extreme or unrealistic values (e.g., negative weight or height).

The program calculates the BMI accurately using the provided weight and height inputs.

Sample test Cases

Test Cases	Test Case1:	Test Case 2:	Test Case 3:	Test Case 4:
Input	Weight: 70.5 kilograms Height: 1.75 meters	Weight: 60.0 kilograms Height: 1.70 meters	Weight: 80.0 kilograms Height: 1.80 meters	Weight: 55.0 kilograms Height: 1.60 meters

Output	BMI: 23.06	BMI: 20.76	BMI: 24.69	BMI: 21.48
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2. Score Message

You are creating a program to display a message for a game score. Ask the user for their score as an integer and then convert it to a string to display a message like "Your score is [score]."

Input format

The program prompts the user to enter their game score as an integer.

Output format

The program converts the integer score to a string and displays a message in the format: "Your score is [score]."

Constraints

The input should be a valid integer representing the game score.

The program converts the input integer to a string and displays it in the specified format.

The program does not handle out-of-range scores or non-integer inputs.

Sample test Cases

Test Cases	Test Case1:	Test Case 2:	Test Case 3:	Test Case 4:
Input	Game Score: 85	Game Score: 120	Game Score: 50	Game Score: 0
Output	Your score is 85.	Your score is 120.	Your score is 50.	Your score is 0.

3.Average Rating

You are developing a program for a movie rating system. Request the user to enter an average rating as a floating-point number, and then convert it to an integer to display the rounded rating.

Input format

The program prompts the user to enter the average movie rating as a floating-point number.

Output format

The program calculates and displays the rounded movie rating as an integer.

Constraints

The input should be a valid floating-point number representing the average movie rating.

The program converts the input to an integer using rounding (nearest integer) without any additional conditions.

The program does not handle out-of-range ratings or non-numeric inputs.

Sample test Cases

Test Cases	Test Case1:	Test Case 2:	Test Case 3:	Test Case 4:
Input	Average Rating: 4.3	Average Rating: 9.8	Average Rating: 6.0	Average Rating: 7.5
Output	Rounded Rating: 4	Rounded Rating: 10	Rounded Rating: 6	Rounded Rating: 8

4. List of Products

You are building an e-commerce application. Ask the user to enter a list of product names as a string, and then convert it to a list of strings to display the products.

Input format

The user is prompted to enter a list of product names as a string.

Product names are separated by commas (,).

Output format

The program displays the list of products, each on a separate line.

Constraints

The input should be a string of product names separated by commas.

There are no specific constraints on the length of the input string.

The program assumes that the input format adheres to the comma-separated list of product names.

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:	Test Case 3:	Test Case 4:
Input	Enter a list of product names separated by commas: Pen, Pencil, Eraser	Enter a list of product names separated by commas: Apples, Oranges, Bananas, Grapes, Kiwi	Enter a list of product names separated by commas: Jacket, Hat, Gloves, Scarf, Boots	Enter a list of product names separated by commas: Chair, Table, Lamp, Rug, Sofa
Output	List of Products: Pen Pencil Eraser	List of Products: Apples Oranges Bananas Grapes Kiwi	List of Products: Jacket Hat Gloves Scarf Boots	List of Products: Chair Table Lamp Rug Sofa

5. Write a program that asks the user to enter a string. The program should then print the following:

- a. The total number of characters in the string**
- b. The string repeated 2 times**
- c. The first character of the string (remember that string indices start at 0)**
- d. The first three characters of the string**

Input format

The first line of each test case consists of String.

Output format

Print the following

- The total number of characters in the string
- The string repeated 2 times
- The first character of the string (remember that string indices start at 0)
- The first three characters of the string

Constraints

$1 \leq S \leq 100$

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:	Test Case 3:	Test Case 4:
String1	hai	hello	welcome	Good
Output	3 haihai h hai	5 hellohello h hell	7 welcomewelcome w welc	4 GoodGood G Good

Level 3: Hard**1.Circle Properties**

Develop a program that prompts the user to enter the radius of a circle and calculates not only the area and circumference but also the diameter, the sector area for a specified angle, and the arc length.give input output format and constraints

Input format

The user is prompted to enter the radius of the circle as a floating-point number.

The user is asked to enter a specified angle in degrees (for sector area) as a floating-point number.

Output format

The program displays the calculated values for radius, diameter, circumference, sector area, and arc length.

Constraints

The radius should be a positive real number (greater than 0).

The angle for sector area should be between 0 and 360 degrees

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:	Test Case 3:	Test Case 4:
Input	Enter the radius of the circle: 5.0 Enter the angle in degrees (for sector area): 60.	Enter the radius of the circle: 3.5 Enter the angle in degrees (for sector area): 120.0	Enter the radius of the circle: 7.0 Enter the angle in degrees (for sector area): 45.0	Enter the radius of the circle: 8.2 Enter the angle in degrees (for sector area): 180.0
Output	Radius: 5.0 Diameter: 10.0 Circumference: 31.41592653589793 Sector Area for 60.0 degrees: 7.853981633974483 Arc Length for 60.0 degrees: 10.471975511965978	Radius: 3.5 Diameter: 7.0 Circumference: 21.991148575128552 Sector Area for 120.0 degrees: 13.107043568319963 Arc Length for 120.0 degrees: 11.596504723931954	Radius: 7.0 Diameter: 14.0 Circumference: 43.982297150257104 Sector Area for 45.0 degrees: 9.817477042468574 Arc Length for 45.0 degrees: 6.578158229168188	Radius: 8.2 Diameter: 16.4 Circumference: 51.513417119032875 Sector Area for 180.0 degrees: 67.09343637467228 Arc Length for 180.0 degrees: 26.67171818733614

2. Total Cost Calculation

You are building a point-of-sale system. Prompt the user to enter the price of an item as a float and the quantity as an integer. Calculate the total cost, which involves implicit conversion between float and integer.

Input format

The price of the item as a floating-point number

The quantity of the item as an integer

Output format

The total cost as a floating-point number with two decimal places

Constraints

Price should be a non-negative floating-point number.

Quantity should be a non-negative integer.

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:	Test Case 3:	Test Case 4:
Input	Enter the price of the item: 7.5 Enter the quantity: 4	Enter the price of the item: 10.25 Enter the quantity: 2	Enter the price of the item: 3.99 Enter the quantity: 1	Enter the price of the item: 0.99 Enter the quantity: 0
Output	Total cost: \$30.00	Total cost: \$20.50	Total cost: \$3.99	Total cost: \$0.00

3. Budget Tracking

You are working on a budgeting app. Request the user to enter their monthly income as a float and a list of expenses as strings. The app will implicitly convert strings to floating-point numbers for calculations.

Input format

Monthly income as a floating-point number (e.g., 2500.75).

List of expenses as strings, separated by spaces.

Output format

The remaining budget as a floating-point number with two decimal places

Constraints

Monthly income should be a non-negative floating-point number.

Each expense should be a non-negative floating-point number.

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:	Test Case 3:	Test Case 4:
Input	Enter your monthly income: 3500.00 Enter your expenses (space-separated): 200.00 150.50 450.75	Enter your monthly income: 500.25 Enter your expenses (space-separated): 30.00 40.50 25.75	Enter your monthly income: 1200.50 Enter your expenses (space-separated): 300.00 450.75 100.25	Enter your monthly income: 800.00 Enter your expenses (space-separated): 150.25 120.50 90.75
Output	Remaining budget: \$2698.75	Remaining budget: \$403.00	Remaining budget: \$349.50	Remaining budget: \$438.50

4. Calculate the Area of a Triangle

Write a Python program to calculate the area of a triangle. The program should prompt the user to enter the lengths of the three sides of the triangle: a, b, and c. After receiving the input, the program should determine whether the input values can form a valid triangle. If the input represents a valid triangle, the program should calculate and display the area of the triangle using Heron's formula. If the input does not represent a valid triangle, the program should display an error message.

Input format

Three positive floating-point numbers representing the lengths of the sides of the triangle (a, b, c).

Output format

If the input values represent a valid triangle, the program should display the calculated area of the triangle as a floating-point number.

If the input does not form a valid triangle (i.e., the sum of any two sides is not greater than the third side), the program should display an error message.

Constraints

The input values should be positive floating-point numbers.

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:	Test Case 3:	Test Case 4:
Input	Enter quantities of products (comma-separated): 10,20,5 Enter prices of products (comma-separated): 2.5,1.99,10.0	Enter quantities of products (comma-separated): 5,15,3,8 Enter prices of products (comma-separated): 1.0,5.0,2.5,3.5	Enter quantities of products (comma-separated): 3,7,2,4 Enter prices of products (comma-separated): 5.25,3.99,12.5,2.75	Enter quantities of products (comma-separated): 10,50,0,15 Enter prices of products (comma-separated): 1.0,0.5,0.0,2.0
Output	Total value of the inventory: 77.50	Total value of the inventory: 104.50	Total value of the inventory: 100.25	Total value of the inventory: 52.50

5. Price Calculator

You are developing a shopping cart program. Prompt the user to enter the price of items as integers and their quantity as integers as well. Calculate the total price, which involves implicit conversion between integers for multiplication.

Input format

For each item, the user is prompted to enter the price as an integer.

For each item, the user is prompted to enter the quantity as an integer.

After entering the price and quantity for an item, the user can choose to enter more items or indicate they are done.

Output format

The program calculates and displays the total price as an integer.

Constraints

The price of each item is a non-negative integer.

The quantity of each item is a non-negative integer.

The total price does not exceed the maximum integer value.

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:	Test Case 3:	Test Case 4:
Input	Enter the price of the first item: 10 Enter the quantity of the first item: 3 Do you want to enter another item? (yes/no): yes Enter the price of the second item: 15 Enter the quantity of the second item: 2 Do you want to enter another item? (yes/no): no	Enter the price of the first item: 5 Enter the quantity of the first item: 10 Do you want to enter another item? (yes/no): no	Enter the price of the first item: 25 Enter the quantity of the first item: 4 Do you want to enter another item? (yes/no): yes Enter the price of the second item: 8 Enter the quantity of the second item: 6	Enter the price of the first item: 12 Enter the quantity of the first item: 5 Do you want to enter another item? (yes/no): yes Enter the price of the second item: 18 Enter the quantity of the

			Do you want to enter another item? (yes/no): yes Enter the price of the third item: 20 Enter the quantity of the third item: 2 Do you want to enter another item? (yes/no): no	second item: 8 Do you want to enter another item? (yes/no): yes Enter the price of the third item: 7 Enter the quantity of the third item: 3 Do you want to enter another item? (yes/no): yes Enter the price of the fourth item: 30 Enter the quantity of the fourth item: 2 Do you want to enter another item? (yes/no): no
Output	Total Price: 45	Total Price: 50	Total Price: 252	Total Price:

				316
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